

GOVERNMENT COLLEGE FOR WOMEN, KARNAL



URL AND PASSWORD SECURITY ANALYZER

*A Project Report on Website URL and
Password Security Analysis*

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DEPARTMENT:

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URL and Password Security Analyzer

1. Title

Cyber Security Project Documentation

2. Project Name

URL and Password Security Analyzer

3. Objective

The objective of this project is to develop a security analysis tool that evaluates website URLs and user passwords for common security risks. The system helps users identify unsafe URLs and weak passwords, thereby improving awareness about cyber threats and promoting secure online practices.

4. Problem Statement

Many internet users unknowingly visit malicious websites or use weak passwords, making them vulnerable to phishing attacks, hacking, and data theft. There is a need for a simple tool that can analyze URLs and passwords, identify security issues, and provide suggestions for improving security.

5. Research Findings

The study identified the following key findings:

- Weak passwords are one of the major causes of unauthorized account access.
- Phishing websites often imitate trusted websites to steal sensitive information.
- Users generally fail to recognize suspicious URLs.
- Passwords with uppercase letters, lowercase letters, numbers, and special characters are significantly more secure.

- HTTPS-enabled websites are generally safer than HTTP websites.
- Real-time security analysis can reduce the chances of cyber attacks.

6. Methodology

The project follows these steps:

1. User enters a website URL.

2. System checks:

- HTTPS availability
- URL length
- Suspicious keywords
- IP address usage

3. User enters a password.

4. Password is analyzed based on:

- Length

- Uppercase letters

- Lowercase letters

- Numbers

- Special characters

5. Security score is generated.

6. Suggestions are displayed to improve password strength and identify unsafe URLs.

7. Tools Used

Programming Language

- Python

Libraries

- Tkinter (GUI)
- re (Regular Expressions)
- urllib
- hashlib

Development Environment

- Visual Studio Code

Operating System

- Windows 10/11

8. Implementation

Module 1: URL Security Analysis

- Accepts website URL
- Detects HTTP/HTTPS
- Checks suspicious keywords
- Identifies abnormal URL structure
- Generates security status

Module 2: Password Strength Analysis

- Calculates password strength
- Checks complexity rules
- Assigns security score

Module 3: Report Generation

- Displays security result
- Shows recommendations
- Helps users improve cybersecurity awareness



1. Home Screen



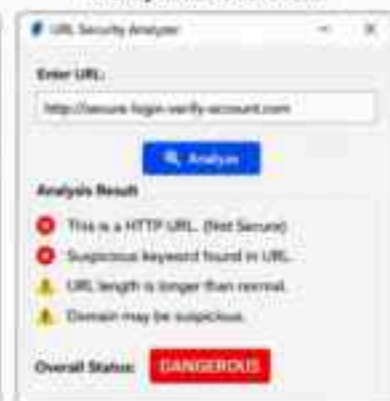
2. URL Analyzer Interface



3. Safe URL Result



4. Suspicious URL Result



5. Password Analyzer Screen



6. Weak Password Result



7. Strong Password Result



9. Screenshots

Include screenshots of:

- Home Screen
- URL Analyzer Interface
- Safe URL Result
- Suspicious URL Result
- Password Analyzer Screen
- Weak Password Result
- Strong Password Result

10. Results

The developed system successfully:

- Detects suspicious website URLs.
- Identifies weak passwords.
- Generates password strength scores.
- Provides security recommendations.
- Improves user awareness regarding cyber security.

The project performed efficiently with minimal processing time and produced accurate security assessments based on predefined rules.

11. Challenges

- Designing an easy-to-use interface.
- Creating accurate URL validation rules.
- Handling different URL formats.
- Generating meaningful password strength scores.
- Reducing false positive security alerts.

12. Future Scope

Future enhancements may include:

- Machine Learning-based phishing detection.
- VirusTotal API integration.
- Real-time website reputation checking.
- Dark Web password breach detection.
- Browser extension support.
- Email phishing detection.
- Mobile application  development.

13. Conclusion

The URL and Password Security

Analyzer is a useful cyber security application designed to improve online safety by analyzing website URLs and password strength. The project effectively identifies potential security risks and provides practical recommendations to users. It serves as an educational as well as practical tool for promoting safe internet usage and demonstrates the application of cyber security concepts in solving real-world problems.



THANK YOU

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